

OpenStax Calculus Volume 3

Section 6.6 Exercises: Surface Integrals

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For the following exercises, determine whether the statements are true or false.

1. If surface S is given by $\{(x, y, z) : 0 \leq x \leq 1, 0 \leq y \leq 1, z = 10\}$, then $\iint_S f(x, y, z) dS = \int_0^1 \int_0^1 f(x, y, 10) dx dy$.
2. If surface S is given by $\{(x, y, z) : 0 \leq x \leq 1, 0 \leq y \leq 1, z = x\}$, then $\iint_S f(x, y, z) dS = \int_0^1 \int_0^1 f(x, y, x) dx dy$.
3. Surface $\mathbf{r} = \langle v \cos u, v \sin u, v^2 \rangle$, for $0 \leq u \leq \pi, 0 \leq v \leq 2$, is the same as surface $\mathbf{r} = \langle \sqrt{v} \cos 2u, \sqrt{v} \sin 2u, v \rangle$, for $0 \leq u \leq \frac{\pi}{2}, 0 \leq v \leq 4$.
4. Given the standard parameterization of a sphere, normal vectors $\mathbf{t}_u \times \mathbf{t}_v$ are outward normal vectors.

For the following exercises, find parametric descriptions for the following surfaces.

5. Plane $3x - 2y + z = 2$
6. Paraboloid $z = x^2 + y^2$, for $0 \leq z \leq 9$.
7. Plane $2x - 4y + 3z = 16$
8. The frustum of cone $z^2 = x^2 + y^2$, for $2 \leq z \leq 8$
9. The portion of cylinder $x^2 + y^2 = 9$ in the first octant, for $0 \leq z \leq 3$
[Figure: A diagram in three dimensions of a section of a cylinder with radius 3. The center of its circular top is (0,0,3). The section exists for x, y, and z between 0 and 3.]
10. A cone with base radius r and height h , where r and h are positive constants

For the following exercises, use a computer algebra system to approximate the area of the following surfaces using a parametric description of the surface.

11. [T] Half cylinder $\{(r, \theta, z) : r = 4, 0 \leq \theta \leq \pi, 0 \leq z \leq 7\}$
12. [T] Plane $z = 10 - x - y$ above square $|x| \leq 2, |y| \leq 2$

For the following exercises, let S be the hemisphere $x^2 + y^2 + z^2 = 4$, with $z \geq 0$, and evaluate each surface integral.

13. $\iint_S z \, dS$
14. $\iint_S (x - 2y) \, dS$
15. $\iint_S (x^2 + y^2)z \, dS$

For the following exercises, evaluate $\iint_S \mathbf{F} \cdot \mathbf{N} \, dS$ for vector field $\mathbf{F}$, where $\mathbf{N}$ is an upward pointing normal vector to surface S .

16. $\mathbf{F}(x, y, z) = x\mathbf{i} + 2y\mathbf{j} - 3z\mathbf{k}$, and S is that part of plane $15x - 12y + 3z = 6$ that lies above unit square $0 \leq x \leq 1, 0 \leq y \leq 1$.
17. $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j}$, and S is hemisphere $z = \sqrt{1 - x^2 - y^2}$.
18. $\mathbf{F}(x, y, z) = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$, and S is the portion of plane $z = y + 1$ that lies inside cylinder $x^2 + y^2 = 1$.
[Figure: A cylinder and an intersecting plane shown in three-dimensions. S is the portion of the plane $z = y + 1$ inside the cylinder $x^2 + y^2 = 1$.]

For the following exercises, approximate the mass of the lamina that has the shape of given surface S . Round to four decimal places.

19. [T] S is surface $z = 4 - x - 2y$, with $z \geq 0, x \geq 0, y \geq 0$; $\rho = x$.
20. [T] S is surface $z = x^2 + y^2$, with $z \leq 1$; $\rho = z$.
21. [T] S is surface $x^2 + y^2 + z^2 = 5$, with $z \geq 1$; $\rho = \theta^2$.
22. Evaluate $\iint_S (y^2z\mathbf{i} + y^3\mathbf{j} + xz\mathbf{k}) \cdot d\mathbf{S}$, where S is the surface of cube $-1 \leq x \leq 1, -1 \leq y \leq 1$, and $0 \leq z \leq 2$. Assume an outward pointing normal.
23. Evaluate surface integral $\iint_S g \, dS$, where $g(x, y, z) = xz + 2x^2 - 3xy$ and S is the portion of plane $2x - 3y + z = 6$ that lies over unit square $R: 0 \leq x \leq 1, 0 \leq y \leq 1$.
24. Evaluate $\iint_S (x^2 + y - z) \, dS$, where S is the surface defined parametrically by $\mathbf{r}(u, v) = (2u + v)\mathbf{i} + (u - 2v)\mathbf{j} + (u + 3v)\mathbf{k}$ for $0 \leq u \leq 1$, and $0 \leq v \leq 2$.
[Figure: A three-dimensional diagram of the given surface, which appears to be a steeply sloped plane stretching through the (x, y) plane.]
25. [T] Evaluate $\iint_S (x - y^2 + z) \, dS$, where S is the surface defined by $\mathbf{r}(u, v) = u^2\mathbf{i} + v\mathbf{j} + u\mathbf{k}$, $0 \leq u \leq 1, 0 \leq v \leq 1$.
[Figure: A three-dimensional diagram of the given surface, which appears to be a curve with edges parallel to the y -axis. It increases in x components and decreases in z components the further it is from the y axis.]
26. [T] Evaluate $\iint_S (x^2 + y^2 - z) \, dS$ where S is the surface defined by $\mathbf{r}(u, v) = u\mathbf{i} - u^2\mathbf{j} + v\mathbf{k}$, $0 \leq u \leq 2, 0 \leq v \leq 1$.
27. Evaluate $\iint_S (x^2 + y^2) \, dS$, where S is the surface of hemisphere $z = \sqrt{1 - x^2 - y^2}$, and above the plane $z = 0$.
28. Evaluate $\iint_S (x^2 + y^2 + z^2) \, dS$, where S is the portion of plane $z = x + 1$ that lies inside cylinder $x^2 + y^2 = 1$.
29. [T] Evaluate $\iint_S x^2z \, dS$, where S is the portion of cone $z^2 = x^2 + y^2$ that lies between planes $z = 1$ and $z = 4$.

[Figure: A diagram of the given upward opening cone in three dimensions. The cone is cut by planes $z=1$ and $z=4$.]

30. [T] Evaluate $\iint_S (xz/y) dS$, where S is the portion of cylinder $x = y^2$ that lies in the first octant between planes $z = 0, z = 5, y = 1$, and $y = 4$.

[Figure: A diagram of the given cylinder in three-dimensions. It is cut by the planes $z=0, z=5, y=1$, and $y=4$.]

31. [T] Evaluate $\iint_S (z + y) dS$, where S is the part of the graph of $z = \sqrt{1 - x^2}$ in the first octant between the xz -plane and plane $y = 3$.

[Figure: A diagram of the given surface in three dimensions in the first octant between the xz -plane and plane $y=3$. The given graph of $z =$ the square root of $(1-x^2)$ stretches down in a concave down curve from along $(0,y,1)$ to along $(1,y,0)$. It looks like a portion of a horizontal cylinder with base along the xz -plane and height along the y axis.]

32. Evaluate $\iint_S xyz dS$ if S is the part of plane $z = x + y$ that lies over the triangular region in the xy -plane with vertices $(0, 0, 0)$, $(1, 0, 0)$, and $(0, 2, 0)$.

33. Find the mass of a lamina of density $\rho(x, y, z) = z$ in the shape of hemisphere $z = (a^2 - x^2 - y^2)^{1/2}$.

34. Compute $\iint_S \mathbf{F} \cdot \mathbf{N} dS$, where $\mathbf{F}(x, y, z) = x\mathbf{i} - 5y\mathbf{j} + 4z\mathbf{k}$ and $\mathbf{N}$ is an outward normal vector of S , where S is the union of two squares $S_1 : x = 0, 0 \leq y \leq 1, 0 \leq z \leq 1$ and $S_2 : z = 1, 0 \leq x \leq 1, 0 \leq y \leq 1$.

[Figure: A diagram in three dimensions. It shows the square formed by the components $x=0, 0 \leq y \leq 1$, and $0 \leq z \leq 1$. It also shows the square formed by the components $z=1, 0 \leq x \leq 1$, and $0 \leq y \leq 1$.]

35. Compute $\iint_S \mathbf{F} \cdot \mathbf{N} dS$, where $\mathbf{F}(x, y, z) = xy\mathbf{i} + z\mathbf{j} + (x + y)\mathbf{k}$ and $\mathbf{N}$ is an upward pointing normal vector S , where S is the triangular region of the plane $x + y + z = 1$ in the first octant.

36. Compute $\iint_S \mathbf{F} \cdot \mathbf{N} dS$, where $\mathbf{F}(x, y, z) = 2yz\mathbf{i} + (\tan^{-1}(xz))\mathbf{j} + e^{xy}\mathbf{k}$ and $\mathbf{N}$ is an outward normal vector of S , where S is the surface of sphere $x^2 + y^2 + z^2 = 1$.

37. Compute $\iint_S \mathbf{F} \cdot \mathbf{N} dS$, where $\mathbf{F}(x, y, z) = xyz\mathbf{i} + xyz\mathbf{j} + xyz\mathbf{k}$ and $\mathbf{N}$ is an outward normal vector S , where S is the surface of the five faces of the unit cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$ missing $z = 0$.

For the following exercises, express the surface integral as an iterated double integral by using a projection on S on the yz -plane.

38. $\iint_S xy^2z^3 dS$; S is the first-octant portion of plane $2x + 3y + 4z = 12$.

39. $\iint_S (x^2 - 2y + z) dS$; S is the portion of the graph of $4x + y = 8$ bounded by the coordinate planes and plane $z = 6$.

For the following exercises, express the surface integral as an iterated double integral by using a projection on S on the xz -plane

40. $\iint_S xy^2z^3 dS$; S is the first-octant portion of plane $2x + 3y + 4z = 12$.

41. $\iint_S (x^2 - 2y + z) dS$; S is the portion of the graph of $4x + y = 8$ bounded by the coordinate planes and plane $z = 6$.

42. Evaluate surface integral $\iint_S yz dS$, where S is the first-octant part of plane $x + y + z = \lambda$, where λ is a positive constant.

43. Evaluate surface integral $\iint_S (x^2z + y^2z) dS$, where S is hemisphere $x^2 + y^2 + z^2 = a^2, z \geq 0$.
44. Evaluate surface integral $\iint_S z dS$, where S is surface $z = \sqrt{x^2 + y^2}, 0 \leq z \leq 2$.
45. Evaluate surface integral $\iint_S x^2yz dS$, where S is the part of plane $z = 1 + 2x + 3y$ that lies above rectangle $0 \leq x \leq 3$ and $0 \leq y \leq 2$.
46. Evaluate surface integral $\iint_S yz dS$, where S is plane $x + y + z = 1$ that lies in the first octant.
47. Evaluate surface integral $\iint_S yz dS$, where S is the part of plane $z = y + 3$ that lies inside cylinder $x^2 + y^2 = 1$.

For the following exercises, use geometric reasoning to evaluate the given surface integrals.

48. $\iint_S \sqrt{x^2 + y^2 + z^2} dS$, where S is surface $x^2 + y^2 + z^2 = 4, z \geq 0$
49. $\iint_S (x\mathbf{i} + y\mathbf{j}) \cdot dS$, where S is surface $x^2 + y^2 = 4, 1 \leq z \leq 3$, oriented with unit normal vectors pointing outward
50. $\iint_S (z\mathbf{k}) \cdot dS$, where S is disc $x^2 + y^2 \leq 9$ on plane $z = 4$, oriented with unit normal vectors pointing upward
51. A lamina has the shape of a portion of sphere $x^2 + y^2 + z^2 = a^2$ that lies within cone $z = \sqrt{x^2 + y^2}$. Let S be the spherical shell centered at the origin with radius a, and let C be the right circular cone with a vertex at the origin and an axis of symmetry that coincides with the z-axis. Determine the mass of the lamina if $\rho(x, y, z) = x^2y^2z$.
[Figure: A diagram in three dimensions. A cone opens upward with point at the origin and an axis of symmetry that coincides with the z-axis. The upper half of a hemisphere with center at the origin opens downward and is cut off by the xy-plane.]
52. A lamina has the shape of a portion of sphere $x^2 + y^2 + z^2 = a^2$ that lies within cone $z = \cot \phi_0 \sqrt{x^2 + y^2}$. Let S be the spherical shell centered at the origin with radius a, and let C be the right circular cone with a vertex at the origin and an axis of symmetry that coincides with the z-axis. Suppose the angle between the sides of the cone and the z-axis is ϕ_0 , with $0 \leq \phi_0 < \frac{\pi}{2}$. Determine the mass of that portion of the shape enclosed in the intersection of S and C. Assume $\rho(x, y, z) = x^2y^2z$.
[Figure: A diagram in three dimensions. A cone opens upward with point at the origin and an axis of symmetry that coincides with the z-axis. The upper half of a hemisphere with center at the origin opens downward and is cut off by the xy-plane.]
53. A paper cup has the shape of an inverted right circular cone of height 6 in. and radius of top 3 in. If the cup is full of water weighing 62.5 lb/ft³, find the total force exerted by the water on the inside surface of the cup.

For the following exercises, the heat flow vector field for conducting objects is $\mathbf{F} = -k\nabla T$, where $T(x, y, z)$ is the temperature in the object and $k > 0$ is a constant that depends on the material. Find the outward flux of F across the following surfaces S for the given temperature distributions and assume $k = 1$.

54. $T(x, y, z) = 100e^{-x-y}$; S consists of the faces of cube $|x| \leq 1, |y| \leq 1, |z| \leq 1$.
55. $T(x, y, z) = -\ln(x^2 + y^2 + z^2)$; S is sphere $x^2 + y^2 + z^2 = a^2$.

For the following exercises, consider the radial fields $\mathbf{F} = \frac{\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{\frac{p}{2}}} = \frac{\mathbf{r}}{\|\mathbf{r}\|^p}$, where p is a real number. Let S consist of spheres A and B centered at the origin with radii $0 < a < b$. The total outward flux across S consists of the outward flux across the outer sphere B less the flux into S across inner sphere A .

[Figure: A diagram in three dimensions of two spheres, one contained completely inside the other. Their centers are both at the origin. Arrows point in toward the origin from outside both spheres.]

56. Find the total flux across S with $p = 0$.
57. Show that for $p = 3$ the flux across S is independent of a and b .